

USN

--	--	--	--	--	--	--	--	--	--

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester M.Tech (MMD) (CBCS) Degree Examinations
 May-June 2025

22MMD201 – THEORY OF VIBRATIONS

Duration: 3 Hours

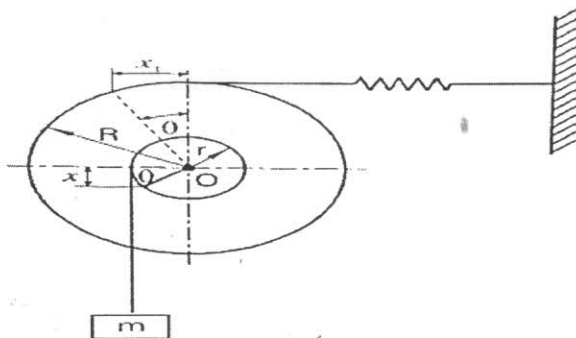
Max. Marks: 100

Note: Answer **Five** full questions choosing **One** full question from **each Unit**.

Unit – I

	Marks	BT*	CO*	PO*
1. a) Add two harmonics: : $x_1 = 15 \sin (\omega t+30^\circ)$ and $x_2 = 8 \cos (\omega t+60^\circ)$.	5	L*2	01	1
b) Obtain a complete solution for a damped free vibration system with damping ratio equal to 1, with the initial displacement of x_0 and zero velocity.	10	L3	01	1
c) Determine ratio of two successive amplitudes for a given system having mass 5 kg, stiffness $k = 500$ N/m and coefficient of damping $C=40$ Ns/m.	5	L3	01	2

2. a) Determine natural frequency for the given system.



b) A body of mass 70 kg is suspended from a spring which deflects 2 cm under the load. It is subjected to a damping effect adjusted to a value of 0.23 times that required for critical damping. Find the natural frequencies of undamped & damped vibration and also ratio of successive amplitudes for damped vibration. If the body is subjected to a periodic disturbing force of 700 N & of frequency equal to 0.78 times the natural frequency. Analyse the amplitude of forced vibration and the phase difference at disturbing force.	5	L3	01	1
c) Obtain an expression for determining the amplitude of vibration in a machine with rotating unbalance.	10	L4	01	2
	5	L2	01	1

Unit – II

3. a) Derive the expression for equation of motion of a SDOF system subjected to undamped step input.	10	L3	02	1
b) How Laplace Transform is used to solve vibration problems?	10	L2	02	1
4. a) Derive the transmissibility of the isolator in case of a vibration isolation system with flexible foundation.	10	L3	02	1
b) An air compressor of mass 200 kg is found to have a large amplitude of vibration, while running at 1200 rpm. Determine the mass and spring constant of the absorber to be added if the natural frequencies of the system are atleast 20% from the impressed frequency.	10	L3	02	2

P.T.O

Unit – III

- | | | | | | |
|-------|---|----|----|----|---|
| 5. a) | Explain the need for measurement of vibration. | 4 | L2 | 03 | 1 |
| b) | Explain the principle behind vibrometer and accelerometer with a neat sketch. | 10 | L2 | 03 | 1 |
| c) | A commercial type of vibration pickup has a natural frequency of 5.7 Hz and damping factor 0.65. What is the lowest frequency beyond which amplitude can be measured with 1% error? | 6 | L3 | 03 | 2 |
| 6. a) | What is phase plane representation? Give the phase plane representation of a simple harmonic oscillator. | 10 | L3 | 03 | 1 |
| b) | Explain the method of isoclinics to construct the trajectories of a dynamical system. | 10 | L2 | 03 | 1 |

Unit – IV

- | | | | | | |
|-------|---|----|----|----|---|
| 7. a) | Explain experimental modal analysis and discuss necessary equipment to carry out the same. | 10 | L2 | 04 | 1 |
| b) | Explain machine condition monitoring techniques and machine maintenance techniques. | 10 | L2 | 04 | 1 |
| 8. a) | A continuous signal has spectral density that is constant $S(f) = 0.004 \text{ cm}^2/\text{Hz}$ between 20 and 1200 Hz. Its mean value is 2 cm. Determine its RMS value and standard deviation. | 10 | L3 | 04 | 1 |
| b) | Write short notes on (i) Power spectrum (ii) Power spectral density (iii) White noise (iv) Wide and narrow band processes. | 10 | L2 | 04 | 1 |

Unit – V

- | | | | | | |
|--------|--|----|----|----|---|
| 9. a) | Derive the free vibration equation for a uniform string. | 10 | L3 | 05 | 1 |
| b) | Derive the equation of free vibration of beams. | 10 | L3 | 05 | 1 |
| 10. a) | Explain the Lanczos method for solving Eigen value problems. | 10 | L2 | 05 | 1 |
| b) | Derive equation of motion for vibration of membranes. | 10 | L3 | 05 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
